

Robstride 03 (RS03) Actuator Specifications for Simulation

Research date: 2026-03-09 **Motor:** Robstride 03 (ROBSTRIDE03), QDD 60 N·m Integrated Joint Motor Module **Manufacturer:** RobStride Dynamics (灵足时代), China

Summary Table

Data sources: "Official spec sheet" = values from Robstride official PDF datasheet (provided directly). "Distributor" = values found in public product listings. Official spec sheet takes priority where there are conflicts.

Parameter	Value	Unit	Source	Notes
Gear ratio	9:1	—	Official spec sheet	Planetary reducer, output:input speed
Rated torque	20	N·m	Official spec sheet	Continuous/rated output torque
Peak torque	60	N·m	Official spec sheet	Short-duration peak at low speed
Rated output power	210	W	Official spec sheet	At rated load (20 N·m × 100 rpm). Note: some distributors list 380 W (see Conflicts section)
No-load speed	195	rpm	Official spec sheet	At 48 V, output shaft
No-load speed	20.42	rad/s	Derived	$195 \text{ rpm} \times 2\pi/60$
Rated load speed	100	rpm	Official spec sheet	Speed at rated torque (20 N·m)
Rated load speed	10.47	rad/s	Derived	$100 \text{ rpm} \times 2\pi/60$
No-load current	0.6	A (RMS)	Official spec sheet	At 48 V
Rated current	12	A (peak)	Official spec sheet	$\approx 8.5 \text{ A RMS}$

Parameter	Value	Unit	Source	Notes
Peak current	43	A (peak)	Official spec sheet	≈ 30.4 A RMS
Torque constant (Kt)	2.36	N·m/A (RMS)	Official spec sheet	Output shaft; includes gear ratio
Motor torque constant (Kt_motor)	~ 0.262	N·m/A	Derived	$Kt_output / gear_ratio = 2.36 / 9$
Back-EMF constant (Ke)	17	V _{rms} /krpm	Official spec sheet	Motor-side back-EMF constant
Line resistance	0.39	Ω	Official spec sheet	Line-to-line; per-phase = $0.195\ \Omega$ (wye)
Winding inductance	0.275	mH	Official spec sheet	Line-to-line
Rated voltage	48	V	Official spec sheet	Optimized; range 15–60 V
Mass	880 ± 20	g	Robstride website	0.880 kg nominal
Torque density	68.18	N·m/kg	Derived	Based on peak torque / 0.880 kg
Pole count	42	poles	Official spec sheet	21 pole pairs
Phases	3	—	Official spec sheet	3-phase BLDC
Drive method	FOC	—	Official spec sheet	Field-oriented control
Encoders	2	—	Distributor	Dual magnetic encoders (14-bit)
Protection rating	IP52	—	Official spec sheet	Customizable to IP67
Operating temperature	–20 to 50	$^{\circ}\text{C}$	Distributor	
Rotor inertia	Not published	kg·m ²	—	See derivation section
Viscous damping	Not published	N·m·s/rad	—	See simulation section

Parameter	Value	Unit	Source	Notes
Coulomb friction	Not published	N·m	—	See simulation section

Physical Dimensions

Parameter	Value	Unit
Outer dimensions	106 × 106 × 56	mm
Weight	880 ± 20	g
Flange hole pattern	30.35 × 9 × 6 (M4)	mm
Mounting holes (perimeter)	8 × M4×8	—
Blind holes (inner ring)	6 × M4×6	—
Output shaft diameter	Ø48 ± 0.2	mm
Inner bore	Ø30.36 ± 0.2	mm

Electrical Interface

Parameter	Value
Power connector	XT30
Signal connector	GH1.25
Voltage range	15–60 V
Rated voltage	48 V

Communication

Parameter	Value
Protocol	CAN (CANopen + MIT mode)
Control frequency	100 Hz (CAN update rate)
Control modes	Position, Velocity, Torque (MIT mode), Current
Position encoding	×1000.0 scale
Velocity encoding	×1000.0 scale
KP encoding	×5.0 scale

Parameter	Value
KD encoding	×500.0 scale
Torque encoding	×10.0 scale

Derived / Calculated Parameters

Motor-side Torque Constant

The Robstride 03 publishes $K_t = 2.36 \text{ N}\cdot\text{m/A}$ as the **output shaft** torque constant (after the 9:1 gearbox). The bare motor's K_t is:

$$K_{t_motor} = K_{t_output} / N = 2.36 / 9 \approx 0.262 \text{ N}\cdot\text{m/A}$$

For a 3-phase BLDC, the back-EMF constant K_e (line-to-neutral, in $\text{V}\cdot\text{s/rad}$) equals K_{t_motor} in SI units. The no-load speed at 48 V gives an estimate:

$$\text{No-load motor speed} = 195 \text{ rpm} \times 9 = 1755 \text{ rpm} = 183.8 \text{ rad/s}$$

$$K_{e_approx} = V_{supply} / \omega_{motor} = 48 / 183.8 \approx 0.261 \text{ V}\cdot\text{s/rad}$$

This is consistent with $K_{t_motor} \approx 0.262 \text{ N}\cdot\text{m/A}$, which is a good cross-check ($K_t = K_e$ in SI for ideal BLDC).

Rated Current Estimate

$$I_{rated} \approx P_{rated} / V_{rated} = 380 \text{ W} / 48 \text{ V} \approx 7.9 \text{ A (RMS)}$$

Check via torque: $I_{rated} = T_{rated} / K_{t_output} = 20 / 2.36 \approx 8.47 \text{ A}$ These are consistent; the ~8 A figure is reasonable for rated continuous operation.

Peak Current Estimate

$$I_{peak} \approx T_{peak} / K_{t_output} = 60 / 2.36 \approx 25.4 \text{ A}$$

This is the estimated peak phase current for producing 60 $\text{N}\cdot\text{m}$ output torque. Not confirmed by a published datasheet.

Rotor Inertia (Bare Motor, Estimated)

No rotor inertia is published. An estimate can be made by assuming a hollow cylindrical rotor geometry:

- Motor outer diameter (stator bore) $\approx 80\text{--}90$ mm (estimated from 106 mm housing, typical for 42-pole pancake BLDC)
- Rotor assumed as thin ring, inner/outer $\approx 35/45$ mm radius, mass $\approx 150\text{--}200$ g (rough estimate)
- $J_{\text{rotor}} \approx m \times r_{\text{avg}}^2 / 2 \approx 0.175 \times 0.040^2 / 2 \approx 1.4 \times 10^{-4} \text{ kg}\cdot\text{m}^2$

This is an order-of-magnitude estimate only. See simulation section for recommended approach.

Known Applications Using Robstride 03

- **K-Bot (K-Scale Labs):** Uses $8 \times$ Robstride 03 (60 N·m) for hip/knee/ankle joints, plus Robstride 04 ($\times 4$), Robstride 02 ($\times 6$), Robstride 00 ($\times 2$). Total 20 actuators.
 - Source: RobStride official X/Twitter, K-Scale Labs GitHub
 - K-Scale's `kscalelabs/kbot`, `kscalelabs/ksim-kbot`, `kscalelabs/kos-sim` repos contain simulation infrastructure but specific MJCF actuator parameters (armature, damping) were not found in public search results.
 - K-Scale also has `kscalelabs/motostandup` with `rotor_robstride04.py`; an equivalent `rotor_robstride03.py` may exist but was not confirmed.
-

Conflicting Values and Uncertainties

Rated Output Power: 210 W vs. 380 W

- **210 W** — from the official Robstride PDF datasheet (confirmed by cross-check: $20 \text{ N}\cdot\text{m} \times 10.47 \text{ rad/s} = 209 \text{ W}$)
- **380 W** — from multiple distributor listings (rcdrone.top, OpenELAB, etc.)

Assessment: 210 W is the correct **rated mechanical output power** at rated load (20 N·m, 100 rpm). The 380 W figure from distributors likely represents either peak power at the maximum-efficiency operating point or input electrical power. Trust the official spec sheet: use 210 W as rated continuous power.

Mass: 880 ± 20 g

- **880 ± 20 g** — Robstride website and most distributor listings (consistent)

Assessment: Use 0.880 kg nominal in simulation. The ± 20 g tolerance can be applied as domain randomization.

No-load Speed: 195 rpm vs. 215 rpm

- **195 rpm** — stated as no-load speed in official spec sheet and distributor listings
- **215 rpm** — described as the speed at which torque "gradually decreases to zero" in some product descriptions

Assessment: 195 rpm is the no-load speed at 48 V. 215 rpm may be at slightly higher voltage (the actuator accepts up to 60 V) or an extrapolated zero-torque intercept from the torque-speed curve.

Torque Constant: Output vs. Motor Side

- **2.36 N·m/A (RMS)** — output shaft K_t (post-gearbox), from official spec sheet. All sources agree.
- Motor-side $K_t = 2.36 / 9 \approx$ **0.262 N·m/A** (derived)

Rotor Inertia

Not published in the official spec sheet or any public source. Must be estimated or measured.

Gear Efficiency

Not published. Typical single-stage planetary: 94–98% efficiency. Expected to be backdrivable (QDD design).

Simulation Parameters

For MuJoCo / Isaac Lab (Mechanical Simulation)

These are the parameters you need for joint simulation. Items marked "estimated" or "recommended" require engineering judgment.

Confirmed Parameters

Parameter	Value	Notes
<code>gear</code> (MuJoCo)	9	Reduction ratio: motor speed = 9 × joint speed
<code>forcerange</code> / torque limit	± 60 N·m	Peak torque at output shaft

Parameter	Value	Notes
<code>ctrlrange</code>	$\pm 60 \text{ N}\cdot\text{m}$	Or $\pm 20 \text{ N}\cdot\text{m}$ for rated continuous
Max joint velocity	20.4 rad/s	195 rpm = 20.42 rad/s at 48 V
Mass	0.880 kg	$\pm 0.020 \text{ kg}$

Armature (Reflected Rotor Inertia)

In MuJoCo, the `armature` attribute adds inertia: $J_{\text{reflected}} = J_{\text{rotor}} \times N^2$

$$J_{\text{reflected}} = J_{\text{rotor}} \times 9^2 = J_{\text{rotor}} \times 81$$

Since J_{rotor} is not published, here is a bracketed estimate:

Source	J_{rotor} estimate	$J_{\text{reflected}} (\times 81)$
Geometric estimate (hollow ring)	$\sim 1.0\text{--}2.0 \times 10^{-4} \text{ kg}\cdot\text{m}^2$	$\sim 8.1\text{--}16.2 \times 10^{-3} \text{ kg}\cdot\text{m}^2$
Comparison: MIT Mini Cheetah motor	$\sim 2.5 \times 10^{-5} \text{ kg}\cdot\text{m}^2$	$\sim 2.0 \times 10^{-3} \text{ kg}\cdot\text{m}^2$ (N=6)
Comparison: Unitree A1 motor	$\sim 3.0 \times 10^{-5} \text{ kg}\cdot\text{m}^2$	$\sim 3.0 \times 10^{-3} \text{ kg}\cdot\text{m}^2$ (N=10)

Recommended starting value: `armature = 0.005` $\text{kg}\cdot\text{m}^2$ (i.e., $J_{\text{reflected}} \approx 5 \times 10^{-3} \text{ kg}\cdot\text{m}^2$) This assumes $J_{\text{rotor}} \approx 6 \times 10^{-5} \text{ kg}\cdot\text{m}^2 \times 81 \approx 5 \times 10^{-3} \text{ kg}\cdot\text{m}^2$. Tune via system identification.

K-Scale Labs reference (K-Bot v2 MJCF — `kscalelabs/ksim-kbot`): The K-Bot v2 MJCF (`kbot-v2-feet/robot.mjcf`) defines separate classes per motor type. For RS03 joints specifically (`motor_03` class, used for shoulder pitch/roll and hip roll/yaw):

```
<default class="motor_03">
  <joint armature="0.005" frictionloss="0.001" actuatorfrange="-100.0 100.0" />
  <motor ctrlrange="-100.0 100.0" />
</default>
```

- `armature = 0.005` $\text{kg}\cdot\text{m}^2$ — identical to the geometric estimate above; this corroborates the 0.005 value
- `frictionloss = 0.001` $\text{N}\cdot\text{m}$ — very low; K-Scale treats RS03 as near-frictionless in sim
- `forcerange =  $\pm 100$` $\text{N}\cdot\text{m}$ — intentionally set above the physical 60 $\text{N}\cdot\text{m}$ peak, giving training headroom; the physical actuator will saturate at 60 $\text{N}\cdot\text{m}$ regardless

Note: the `walking.py` task config uses a high-level `armature = 1e-2` default that does NOT override per-class MJCF values — the MJCF `motor_03` class value of 0.005 is what's actually applied to RS03 joints.

Viscous Damping (joint-level)

Not published. In MuJoCo, `damping` is in $\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$ at the joint (output) shaft.

$$b_{\text{joint}} = b_{\text{motor}} \times N^2$$

For QDD actuators with FOC, electrical damping is often used in lieu of mechanical. At the output shaft, a typical starting value for a 60 $\text{N}\cdot\text{m}$ actuator:

Recommended: `damping = 1.0` $\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$ (output shaft) This is a common starting point for human-scale QDD actuators; tune via pendulum drop test.

Coulomb / Static Friction

Not published. For a 9:1 planetary gearbox: - Planetary gear stages typically have low Coulomb friction (backdrivable design) - Estimate: 0.5–2.0% of rated torque at output

Recommended: `frictionloss = 0.5` $\text{N}\cdot\text{m}$ (output shaft) as a starting point

Motor Torque Constant (for electrical simulation or torque control)

$$K_{t_output} = 2.36 \text{ N}\cdot\text{m}/\text{A} \text{ (RMS)}$$

$$K_{t_motor} = 0.262 \text{ N}\cdot\text{m}/\text{A} \text{ (bare motor)}$$

For Isaac Lab `ImplicitActuatorCfg` or `DCMotorCfg`:

$$\text{effort_limit} = 60.0 \quad \# \text{ N}\cdot\text{m} \text{ (peak)}$$

$$\text{velocity_limit} = 20.4 \quad \# \text{ rad/s (195 rpm)}$$

$$\text{stiffness} = \{\dots\} \quad \# \text{ PD gains (task-dependent)}$$

$$\text{damping} = \{\dots\} \quad \# \text{ PD gains (task-dependent)}$$

For `DCMotorCfg` (if using actuator net or motor model):

$$\text{motor_torque_constant} = 0.262 \quad \# \text{ N}\cdot\text{m}/\text{A} \text{ (motor side)}$$

$$\text{gear_ratio} = 9.0$$

MuJoCo Actuator XML Example

```
<!-- Position servo with motor model -->
```

```
<actuator>
```

```
  <motor name="joint_motor"
```

```
    joint="joint_name"
```

```
    gear="9"
```

```
    forcerange="-60 60"
```

```
    ctrllimited="true"
```

```
    ctrlrange="-60 60"/>
```



```

</actuator>

<!-- Joint properties -->
<joint name="joint_name"
    armature="0.005"
    damping="1.0"
    frictionloss="0.5"/>

```

Isaac Lab `ImplicitActuatorCfg` Example

```

from isaacsim.actuators import ImplicitActuatorCfg

actuators = {
    "joint_name": ImplicitActuatorCfg(
        joint_names_expr=["joint_name"],
        effort_limit=60.0,          # N·m peak
        velocity_limit=20.4,        # rad/s (195 rpm)
        stiffness=80.0,             # Tune for your task
        damping=2.0,                # Tune for your task
        # armature set in USD/URDF articulation config
    ),
}

```

Sources

All data points are sourced below. Parameters not found in public sources are marked in the table as "Not published."

Primary / Manufacturer Sources

1. **RobStride Official Product Page — RS03** <https://robstride.com/products/robStride03> *Peak torque (60 N·m), rated torque (20 N·m), gear ratio (9:1), no-load speed (195 rpm), power (380 W), mass (880 g), IP52, pole count (42), FOC drive, dual encoders*
2. **RobStride GitHub — Product Information** https://github.com/RobStride/Product_Information *Official documentation repository; contains manuals and datasheets (direct file access not retrieved)*
3. **RobStride GitHub — MotorStudio** <https://github.com/RobStride/MotorStudio> *Motor configuration and control software*

4. **ROBSTRIDE03 Integrated Joint Motor Module User Manual (manuals.plus)** <https://manuals.plus/ae/1005007705574359> *Torque-speed curve description (peak 60 N·m, falls toward 215 rpm), operating conditions, voltage range 15–60 V, FOC control, IP52*
5. **Robostride 03 Motor Installation Drawing and Specifications (manuals.plus)** <https://manuals.plus/m/fcfb166db8e37bb6a15a8984f3d6c4bd1534bd174127855f1633084b7408cf48> *Mechanical dimensions: Ø48 ±0.2 output, Ø30.36 ±0.2 inner bore, 54.1 mm length, mounting hole patterns*

Distributor / Retailer Sources

1. **RCDrone — Robstride 03 QDD 60N·m Product Listing** <https://rcdrone.top/products/robstride-03-qdd-60n-m-integrated-actuator-module-48v-dual-encoders-planetary-reducer-ip52-9-1-ratio> *Torque constant 2.36 N·m/Arms, no-load current 0.6 A, no-load speed 195 rpm, voltage range 15–60 V, rated power 380 W, mass 880 g, gear ratio 9:1, IP52, 42 poles, 3-phase FOC*
2. **AIFITLAB — ROBSTRIDE 03 Motor** <https://aifitlab.com/products/robstride-03-motor> *Peak torque 60 N·m, rated torque 20 N·m, gear ratio 9:1, mass 880 g, IP52, 42 poles*
3. **OpenELAB — ROBSTRIDE03 QDD 60N.m** <https://openelab.io/products/robstride03-qdd-60n-m-integrated-joint-motor-module> *Consistent specs: peak 60 N·m, rated 20 N·m, 380 W, 9:1, 880 g*
4. **Seeed Studio — Robostride 00 (reference model)** <https://www.seeedstudio.com/Robostride-00-Actuator-p-6664.html> *Cross-reference for product line consistency*
5. **Amazon — ROBSTRIDE03 (product listing)** <https://www.amazon.com/Articulated-Motor-Module-ROBSTRIDE03-Quasi-Direct/dp/B0FGPYRXRR> *Confirms 60 N·m peak, QDD integrated module*

Control / SDK Documentation

1. **Seeed Studio Wiki — RobStride Motor Control Complete Guide** https://wiki.seeedstudio.com/robstride_control/ *CAN protocol details: MIT mode, position/velocity/torque control; encoding scales (position ×1000, velocity ×1000, kp ×5, kd ×500, torque ×10); 100 Hz control rate; CAN frame structure; damping behavior note*
2. **Rust crate — robstride (actuator_types.rs)** https://docs.rs/robstride/latest/src/robstride/actuator_types.rs.html *Open-source Rust SDK for Robstride actuators; contains Type03 enum definitions; specific numerical limits not retrieved*
3. **Rust crate — robstride (crates.io)** <https://crates.io/crates/robstride> *Robstride Python/Rust SDK*
4. **GitHub — sirwart/robstride (Python SDK)** <https://github.com/sirwart/robstride> *Community Python SDK for Robstride actuators*

K-Scale Labs (K-Bot Application)

1. **RobStride on X (Twitter) — K-Bot announcement** https://x.com/RobStride_com/status/1894292440468598843 *Confirms K-Bot uses $8 \times RS03$ (60 N·m), $4 \times RS04$, $6 \times RS02$, $2 \times RS00$ — 20 total actuators*
2. **K-Scale Labs — kbot GitHub repository** <https://github.com/kscalelabs/kbot> *K-Bot robot design, URDF/MJCF files (direct file access not retrieved)*
3. **K-Scale Labs — ksim-kbot** <https://github.com/kscalelabs/ksim-kbot> *RL training and simulation for K-Bot with MuJoCo; uses Robstride actuators*
4. **K-Scale Labs — kos-sim** <https://github.com/kscalelabs/kos-sim> *KOS simulation backend*
5. **K-Scale Labs — motostandup (rotor_robstride04.py)** https://github.com/kscalelabs/motostandup/blob/master/rotor_robstride04.py *Contains rotor identification parameters for RS04; RS03 equivalent not found publicly*
6. **DeepWiki — kscalelabs/kbot** <https://deepwiki.com/kscalelabs/kbot> *Wiki-style documentation of K-Bot software architecture*

Academic / Background

1. **Extended Friction Models for the Physics Simulation of Servo Actuators (arXiv 2410.08650)** <https://arxiv.org/abs/2410.08650> *Best available reference for servo actuator friction modeling methodology; proposes pendulum-based system identification for Coulomb + viscous + Stribeck models*
2. **Impact of Static Friction on Sim2Real in Robotic RL (arXiv 2503.01255)** <https://arxiv.org/abs/2503.01255> *Discusses unexpectedly high friction ratios in robotic joints; relevant for sim-to-real gap analysis*

What Was NOT Found

The following parameters are **not in any public or official source** as of 2026-03-09:

Parameter	Status
Rotor inertia (kg·m ²)	Not in any datasheet, manual, or GitHub config — must estimate or measure
Viscous damping coefficient (N·m·s/rad)	Not published — measure via pendulum drop test
	Not published — measure via pendulum drop test

Parameter	Status
Coulomb/static friction torque (N·m)	
Gear efficiency (%)	Not published

The following were found in the **official spec sheet** (not in public web sources):

Parameter	Value
Line resistance	0.39 Ω (line-to-line)
Winding inductance	0.275 mH
Back-EMF constant	17 V _{rms} /krpm
Rated current	13 A peak
Peak current	43 A peak
Rated output power	210 W
Rated load speed	100 rpm

Recommendation: To obtain rotor inertia and friction parameters: 1. Contact RobStride directly (sales@robstride.com or via robstride.com) 2. Perform a **pendulum drop test**: mount the actuator as a pendulum, release from known angle, record oscillation decay, and fit a Coulomb + viscous friction model (see arXiv 2410.08650 for methodology) 3. **Measure phase resistance** with a multimeter (line-to-line; divide by 2 for per-phase resistance) 4. Check the RobStride Product_Information GitHub repo for any PDF datasheets not indexed by search engines 5. Check kscalelabs/kbot and kscalelabs/ksim-kbot for embedded MJCF actuator parameters — these repos were confirmed to use RS03 but direct file access was not available during this research session

Quick-Reference for Isaac Lab Config

```
# Robstride 03 – confirmed parameters
ROBSTRIDE03_EFFORT_LIMIT = 60.0 # N·m (peak output torque)
ROBSTRIDE03_VELOCITY_LIMIT = 20.4 # rad/s (195 rpm no-load @ 48V)
ROBSTRIDE03_GEAR_RATIO = 9.0 # reduction ratio
ROBSTRIDE03_KT_OUTPUT = 2.36 # N·m/Arms (output shaft)
ROBSTRIDE03_KT_MOTOR = 0.262 # N·m/Arms (motor shaft, derived)
ROBSTRIDE03_MASS = 0.880 # kg ( $\pm 0.020$  kg tolerance)
ROBSTRIDE03_RATED_CURRENT = 12.0 # A peak ( $\approx 8.5$  A RMS)
```

```
ROBSTRIDE03_PEAK_CURRENT = 43.0 # A peak ( $\approx 30.4$  A RMS)
```

```
# Estimated simulation parameters (TUNE THESE via motor characterization)
```

```
ROBSTRIDE03_ARMATURE = 0.005 # kg·m2 reflected inertia
```

```
# Corroborated: K-Scale K-Bot v2 MJCF uses 0.005 for RS03
```

```
ROBSTRIDE03_DAMPING = 1.0 # N·m·s/rad viscous (estimated; tune via characterization)
```

```
ROBSTRIDE03_FRICTION = 0.5 # N·m Coulomb (estimated; K-Scale uses 0.001 – very low)
```

```
# K-Scale frictionloss=0.001 suggests RS03 is near-backdriven
```